

DS 310 — Homework 1

Database Design

Grading: 57 points total.

Scenario

A start-up Insighta is building a platform that streamlines how researchers recruit participants. Researchers post their study details and eligibility criteria, and only participants who meet those criteria are able to enroll.

Users each have a unique user ID, an email address, a password, and a role (researcher, participant, or both). A user may have multiple phone numbers on file.

Studies are posted by exactly one researcher (a user). Each study has a unique study ID, a title, a type (survey or interview), a status (draft, active, or closed), an estimated duration in minutes, a reward amount, and the date it was published.

Screening Questions belong to exactly one study. Each question has a unique question ID, question text, and an order number. A study may have zero or more screening questions.

Applications represent a participant (a user) applying to a study. The platform records when the application was submitted and its status (pending, approved, completed, or withdrawn). A given participant may apply to the same study at most once.

Part 1 — ER Diagram (10 points)

Construct a reasonable ER diagram for the scenario above.

Q1.1 — Entity sets, attributes, and keys (5 points)

List all entity sets you will include in your ER diagram. For each one, list its attributes and clearly indicate the keys.

Q1.2 — Draw the ER diagram (5 points)

Draw the ER diagram for your design. Show the entity sets, their attributes, the relationships among them, the key attributes, and the cardinality and participation constraints.

Part 2 — Relational Schema (10 points)

Translate your ER diagram into a relational database schema.

Q2.1 — Write the full schema (5 points)

Write the complete schema using the form `RelationName(attr1, attr2, attr3, ...)`. Follow these rules:

- Underline primary key attributes.
- Annotate foreign keys, e.g., `study_id → Study(study_id)`.
- Handle multi-valued attributes correctly (each requires its own relation).

- Apply the many-to-one optimization where appropriate, and state when you do so.

Q2.2 — Reasoning about NULLs (5 points)

A researcher just created a brand-new study but has not yet published it (`status = 'draft'`). No participants have applied. For each relation in your schema, state whether this study's data appears in it, and if so, whether any attributes would be NULL. Explain your reasoning for any relation where NULLs appear or where the study is absent entirely.

Part 3 — Keys and Constraints (12 points)

Q3.1 — Keys vs. candidate keys (12 points)

Consider the following new relation:

```
ScreeningAnswer(application_id, question_id, answer_text)
```

Assume each application answers each screening question *at most once*.

For each attribute set below, state whether it is a key, a candidate key, both, or neither, and justify in one sentence.

Attribute set	Key?	Candidate key?	Justification
<code>(application_id, question_id)</code>			
<code>application_id</code>			
<code>(application_id, question_id, answer_text)</code>			
<code>answer_text</code>			

Part 4 — Relational Algebra (25 points)

Use the schema you defined in Part 2. For each query, write a relational algebra expression. Use the operators introduced in class, for example: σ , π , $\times$, $\bowtie$, ρ , $-$.

Q4.1 (5 points)

Find the titles of all studies with a reward amount greater than \$10.

Q4.2 (5 points)

Find the email addresses of all users with role `'researcher'`.

Q4.3 (5 points)

Find the titles of all studies that the participant with `user_id = 42` has completed.

Q4.4 (5 points)

Find the titles and reward amounts of all active studies for which participant 42 has not yet applied.

Q4.5 (5 points)

Find the titles of all studies that have received at least one application with status 'completed'.

Submitting

Submit your completed write-up, including your ER diagram, as a PDF to Gradescope.